

Yu Chen (Terri) Tai

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EDUCATION

University of California, San Diego

San Diego, CA

B.S. in Computer Engineering GPA: 3.87/4.0

Expected June 2027

- **Honors:** IEEE-Eta Kappa Nu honor society; 13th in U.S., IEEE Xtreme Programming Competition 2024; 5th place, WIC Beginner Programming Competition 2024; Stanford TreeHacks 2025 (8% acceptance rate)
- **Coursework:** Data Structures & OOD, Operating Systems, Machine Learning, Autonomous Vehicles, Computer Organization, Probability & Statistics, Linear Algebra, Differential Equations

TECHNICAL EXPERIENCE

Software Development Engineer Intern

Jun 2026 – Sep 2026

AWS Cryptography — Secure Channel Systems

Seattle, WA

- Designed and built a **Java** service for an internal **Public Key Infrastructure** (PKI) that reads allowlisting configuration from **AWS AppConfig**, replacing hardcoded service-principals list with a dynamic configuration.
- Built and shipped a dedicated **distributed deployment pipeline** propagating configuration changes across all customer cells in every AWS region, cutting rollout time from **2 weeks to 3–4 hours** (>95%).
- Built with **Java** and **TypeScript** for encrypted, mutually authenticated service communication across a multi-region infrastructure system.

Software Team Lead — RoboCup Soccer

Oct 2023 – Dec 2025

IEEE UCSD

San Diego, CA

- Lead a 25-person engineering team building autonomous soccer robots in **Java**, applying **object-oriented design** and **data structures & algorithms**; run **Agile/SCRUM** sprints and code reviews across the full **SDLC**.
- Redesigned the core obstacle avoidance module by replacing a legacy shape-based system with an **A* path planning algorithm**, opening up **75%** more navigable space and route flexibility.
- Built a vector-based dribbling function that finds the optimal direction, improving scoring efficiency by **15%**.

Undergraduate Researcher

Jun 2025 – Aug 2025

Existential Robotics Lab (ERL) — GEAR Program, UC San Diego

La Jolla, CA

- Implemented **Particle Filter SLAM** and **Kalman Filter**-based motion prediction for a multi-agent **TurtleBot4** system in **ROS2** using **Python/C++**.
- Conducted robotics **research** under Professor Nikolay Atanasov, validating estimation algorithms in **Gazebo** simulation before hardware deployment.

CSE Tutor — CSE 11 & CSE 12

Sep 2024 – Present

UCSD Computer Science & Engineering Department

San Diego, CA

- Hold weekly office hours for **30+** students covering introductory **Java**, **data structures**, and debugging strategies.
- Collaborated with TAs and tutors to design assignments, review sessions, and exam materials for **100+** students.

PROJECTS

Brain Tumor Segmentation (GitHub) | Python, PyTorch

Jan 2026 – Mar 2026

- Implemented a **2D U-Net CNN from scratch** in **PyTorch** to train a **machine learning** model for automated segmentation on 4-channel multimodal MRI scans using the **BraTS** dataset.
- Wrote a custom **Dice Loss** function to handle class imbalance; tracked performance via Dice coefficient.

WayNo — Autonomous Parallel Parking Robot (GitHub) | Python, ROS2, OpenCV

Sep 2025 – Dec 2025

- Designed and shipped a real-time parallel parking algorithm combining **sensor fusion** and **path planning**; achieved **>95% accuracy** on physical hardware in a team of 4.
- Built a computer vision pipeline using **HSV color masking** and contour detection for reliable sign recognition.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, HTML/CSS, Go, Swift, ARM32 Assembly

Frameworks & Tools: ROS2, PyTorch, OpenCV, Git, Linux/Ubuntu, Firebase, MATLAB

Concepts: Data Structures & Algorithms, OOP/OOD, Machine Learning, Distributed Systems, Agile/SCRUM, SLAM

AI Tooling: GitHub Copilot, Claude Code, ChatGPT (GenAI-assisted development)